SYLLABUS : ELECTROMAGNETIC FORCE

1. A circular loop of area 1 cm^2, carrying a current of 10 A, is placed in a magnetic field of 0.1 T perpendicular to the plane of the loop. The torque on the loop due to the magnetic field is

(A) zero (B) 10^{-4} N-m (C) 10^{-2} N-m (D) 1 N-m

2. A particle moves in a circle of radius 1.0 cm under the action of a magnetic field of 0.40 T. An electric field of 200 V/m makes the path straight. Find the charge/mass ratio of the particle.

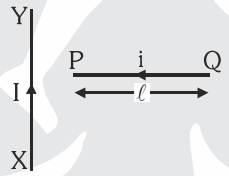
(A) $\frac{3}{4} \times 10^5$ C/kg (B) $\frac{5}{4} \times 10^5$ C/kg (C) $\frac{5}{8} \times 10^5$ C/kg (D) $\frac{5}{3} \times 10^5$ C/kg

3. A particle having mass m and charge q is released from the origin in a region in which electric field and magnetic field are given by $\vec{B} = +B_0\hat{j}$ and $\vec{E} = +E_0\hat{i}$.

Find the speed of the particle as a function of its X-coordinate.

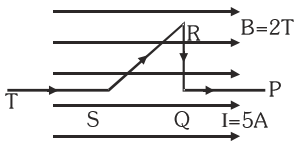
(A) $\sqrt{\frac{3qE_0x}{m}}$ (B) $\sqrt{\frac{4qE_0x}{m}}$ (C) $\sqrt{\frac{qE_0x}{m}}$ (D) $\sqrt{\frac{2qE_0x}{m}}$

4. A conductor PQ carries a current 'i' is placed perpendicular to a long conductor XY carrying a current I. The direction of force on PQ will be :–



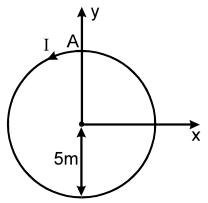
(A) towards right (B) towards left (C) upwards (D) downwards

5. A wire PQRST carrying current I = 5A is placed in uniform magnetic field B = 2T as shown in fig. If the length of part QR = 4 cm and SR = 6 cm then the magnetic force on SR edge of the wire is :–



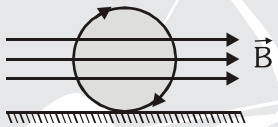
(A) 0.4 N (B) 0.6 N (C) zero (D) 6 N

6. A ring of radius 5 m is lying in the x-y plane and is carrying current of 1 A in anti-clockwise sense. If a uniform magnetic field $\vec{B} = 3\hat{i} + 4\hat{j}$ is switched on, then the co-ordinates of point about which the loop will lift up is:



(A) (3, 4) (B) (4, 3) (C) (3, 0) (D) (0, 3)

7. A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane. The ring carries a current i = 4A. A horizontal magnetic field B = 10 T is switched on at time t = 0 as shown in figure. The initial angular acceleration of the ring will be



(A) 40π rad/s^2 (B) 20π rad/s^2 (C) 5π rad/s^2 (D) 15π rad/s^2

8. A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60º. The torque needed to maintain the needle in this position will be :

(A) $\sqrt{3}$ W (B) W (C) $(\sqrt{3}/2)$W (D) 2 W

9. There are four light–weight–rod samples A,B,C,D separately suspended by threads. A bar magnet is slowly brought near each sample and the following observations are noted :

(i) A is feebly repelled (ii) B is feebly attracted

(iii) C is strongly attracted (iv) D remains unaffected

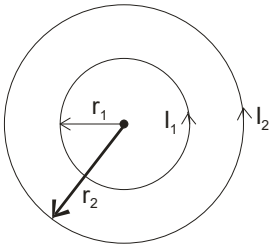
Which one of the following is true ?

(A) B is of a paramagnetic material (B) C is of a diamagnetic material

(C) D is of a ferromagnetic material (D) A is of a non–magnetic material

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10. A and B are two concentric circular conductors of centre O and carrying current i_1 and i_2 as shown in the diagram. If ratio of their radii is 1:2 and ratio of the flux densities at O due to A and B is 1 : 3 then the value of $\frac{i_1}{i_2}$ will be :–



(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{6}$

11. Two thick wires and two thin wires, all of the same materials and same length form a square in the three different ways P, Q and R as shown in fig with current direction shown, the magnetic field at the centre of the square is zero in cases



(A) in P only (B) in P and Q only (C) in Q and R only (D) P and R only

12. A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60º. The torque needed to maintain the needle in this position will be :

(A) $\sqrt{3}\,W$ (B) W (C) $(\sqrt{3}/2)W$ (D) 2 W

13. A magnetic lines of force inside a bar magnet :

(A) are from north-pole to south-pole of the magnet

(B) do not exist

(C) depend upon the area of cross-section of the bar magnet

(D) are from south-pole to north-pole of the magnet

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14. A current i ampere flows along an infinitely long straight thin walled tube, then the magnetic induction at any point inside the tube is :

(A) infinite (B) zero (C) $\frac{\mu_0}{4\pi}, \frac{2i}{r}$ tesla (D) $\frac{2i}{r}$ tesla

15. Two long conductors, separated by a distance d carry currents I_1 and I_2 in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to 3d. The new value of the force between them is :

(A) –2 F (B) F/3 (C) –2F/3 (D) – F/3

16. A current i ampere flows along an infinitely long straight thin walled tube, then the magnetic induction at any point inside the tube is :

(A) infinite (B) zero (C) $\frac{\mu_0}{4\pi}, \frac{2i}{r}$ tesla (D) $\frac{2i}{r}$ tesla

17. Two thin, long, parallel wires, separated by a distance 'd' carry a current of 'i' A in the same direction. They will :

(A) attract each other with a force of $\frac{\mu_0 i^2}{(2\pi d)}$ (B) repel, each other with a force of $\frac{\mu_0 i^2}{(2\pi d)}$

(C) attract each other with a force of $\frac{\mu_0 i^2}{(2\pi d^2)}$ (D) repel each other with a force of $\frac{\mu_0 i^2}{(2\pi d^2)}$

18. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected along the direction of the fields with a certain velocity, then

(A) its velocity will decrease

(B) its velocity will increase

(C) it will turn towards right of direction of motion

(D) it will turn towards left of direction of motion.

19. A magnetic needle is kept in a non-uniform magnetic field. It experiences :

(A) a torque but not a force (B) neither a force nor a torque

(C) a force and a torque (D) a force but not a torque

20. In a region, steady and uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in this region. The path of the particle will be a :

(A) circle (B) helix (C) straight line (D) ellipse

21. Needles N_1, N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will :

(A) attract all three of them

(B) attract N_1 and N_2 strongly but repel N_3

(C) attract N_1 strongly, N_2 weakly and repel N_3 weakly

(D) attract N_1 strongly, but repel N_2 and N_3 weakly

22. A long solenoid has 200 turns per cm and carries a current i. The magnetic field at its centre is 6.28×10^{-2} Weber/m^2. Another long solenoid has 100 turns per cm and it carries a current i/3. The value of the magnetic field at its centre is :

(A) 1.05×10^{-4} Weber/m^2 (B) 1.05×10^{-2} Weber/m^2

(C) 1.05×10^{-5} Weber/m^2 (D) 1.05×1010^{-3} Weber/m^2

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23. A long straight wire of radius *a* carries a steady current *i*. The current is uniformly distributed across its crosssection. The ratio of the magnetic field at $\frac{a}{2}$ and 2a from axis is :

(A) 1/4 (B) 4 (C) 1 (D) 1/2

24. A current I flows along the length of an infinitely long, straight, thin walled pipe. Then :

(A) the magnetic field is zero only on the axis of the pipe

(B) the magnetic field is different at different points inside the pipe

(C) the magnetic field at any point inside the pipe is zero

(D) the magnetic field at all points inside the pipe is the same, but not zero

25. A charged particle with charge q enters a region of constant, uniform and mutually orthogo nal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then :

(A) $\vec{v}=\vec{E}\times\vec{B}/B^2$ (B) $\vec{v}=\vec{E}\times\vec{E}/B^2$ (C) $\vec{v}=\vec{E}\times\vec{E}/E^2$ (D) $\vec{v}=\vec{B}\times\vec{E}/E^2$

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ANSWER KEY

1.	(A)	**2.**	(B)	**3.**	(D)	**4.**	(D)	**5.**	(A)
6.	(A)	**7.**	(A)	**8.**	(A)	**9.**	(A)	**10.**	(D)
11.	(D)	**12.**	(A)	**13.**	(D)	**14.**	(B)	**15.**	(C)
16.	(B)	**17.**	(A)	**18.**	(A)	**19.**	(C)	**20.**	(C)
21.	(C)	**22.**	(B)	**23.**	(C)	**24.**	(C)	**25.**	(A)